

The Indian roadscape: A stress test for AI

Compared with the relatively organised traffic conditions found in many developed nations, Indian roads represent a significant stress test for AI. Lane discipline is often inconsistent, road quality varies significantly, and unexpected elements, such as cows, stray dogs, or roadside vendors, can appear at any time. Seasonal extremes, including monsoon-driven flooding and poor visibility, add to the difficulty, making India one of the most challenging environments for autonomous driving.

As a result, AI systems deployed in India cannot simply replicate solutions developed elsewhere. They must be localised, trained on Indian datasets, and designed to respond to highly unpredictable human and environmental behaviour.

The role of AI models and edge computing

AI forms the backbone of autonomous navigation, enabling vehicles to process large volumes of sensory data in real time. In India, where road infrastructure remains inconsistent, edge AI plays a critical role. Deep learning algorithms are central to recognising and responding to complex traffic scenarios, including the ability to distinguish between pedestrians, street vendors, and stray animals.

The capacity of AI systems to learn continuously and adapt to local conditions is essential for safe and efficient autonomous driving. Reinforcement learning models, which allow systems to improve through trial and error, are being explored to address the unpredictability of Indian traffic patterns. Running AI models on embedded edge devices such as smartphones, Jetson modules, or vehicle electronic control units (ECUs) reduces reliance on cloud connectivity and enables instant decision-making, even in areas with limited network coverage.

Connectivity, 5G, and V2X ecosystems

The next major advance is expected to come from vehicle-to-everything (V2X) communication, which enables vehicles to communicate with one another, traffic signals, and pedestrians. On Indian roads, V2X has the potential to improve traffic coordination by supporting real-time alerts related to congestion, accidents, and roadblocks.

In densely populated urban environments characterised by unpredictable behaviour, V2X technologies could play a key role in improving traffic flow. Vehicles equipped with V2X systems can receive timely information about incidents or disruptions and adjust routes accordingly. Integration with smart city infrastructure may further enhance coordination between traffic signals, public transport systems, and emergency response services.

Startups such as Vehiqilla are already developing secure V2X platforms to strengthen vehicles' environment communication. However, widespread adoption will require substantial upgrades to road infrastructure, including consistent lane markings, reliable real-time data systems, and intelligent traffic management frameworks. Cybersecurity also remains a critical concern, as V2X systems must safeguard data integrity and prevent malicious interference.

With the rollout of 5G networks across India, ultra-low-latency communication is expected to significantly enhance the reliability of V2X systems. A tightly connected ecosystem, linking autonomous buses, emergency vehicles, and traffic signals, could help reduce congestion while improving overall road safety.

Overcoming public perception and regulatory barriers

Public perception remains a significant barrier to the adoption of autonomous vehicles in India. Many Indians are sceptical about the safety and reliability of driverless

cars, while concerns about job losses in the transport sector add to this hesitation. Building trust through education, transparent trials, and demonstration projects will be critical, particularly if these efforts clearly demonstrate how autonomous systems can reduce accidents and save lives.

On the regulatory front, India's legal framework for autonomous vehicles is still in its infancy. While the government has expressed interest in promoting electric and autonomous vehicles as part of smart city initiatives, comprehensive policies addressing safety, liability, and ethical considerations remain necessary to support widespread adoption. Regulations that ensure robust testing standards and clear certification processes will play a pivotal role in creating a reliable and trusted ecosystem for autonomous driving.

Indian ecosystem: Startups, industry, and policy

The autonomous ecosystem in India is gaining steady momentum, driven by a combination of homegrown startups, established automotive manufacturers, and a deep IT talent base. Rather than pursuing a single moonshot, progress is emerging through tightly scoped deployments in industrial corridors, ports, campuses, and logistics hubs, where autonomy can demonstrate safety and return on investment early.

Among startups, Bengaluru-based Minus Zero is developing an end-to-end ADAS and autonomous driving stack tailored to India's unstructured traffic conditions and cost constraints. Ati Motors, meanwhile, focuses on rugged autonomous mobile robots designed for use in factories and warehouses.

RoshAi, a Kochi-based startup, is entering the space with a full-stack autonomy approach for both on-road and off-road applications. Its portfolio includes a modular, retrofit-ready Level-4 autonomy stack for trucks, buses, and passenger